MILITARY SPECIFICATION
SOLENOID, FIRING, D7665749 and F8208586

1. SCOPE

1.1 This specification covers a 24 volt d-c firing solenoid used on Gun, Twin, 40mm Self-propelled, M42 and a 208 volt, 60 cycle Firing Solenoid used on Carriage, Gun, 40mm M2A3.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and drawings, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

U. S. Army	
51-7	Instruments, Fire Control, General Specification Governing the Manufacture and Inspection of

DRAWINGS

Ordnance Corps	
D7665749	Solenoid Assembly
F8208586	Solenoid

(Copies of specifications, standards and drawings required by the contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General.- Unless otherwise specified, the requirements common to all fire control instruments as contained in Specification 51-7 shall be mandatory.

3.2 Material.- Material shall be in accordance with the applicable drawings and specifications forming a part of this specification.

3.3 Electrical

3.3.1 Rated voltage

3.3.1.1 Solenoid assembly D7665749. - Each solenoid shall meet the performance requirements of this specification when 26.5 volts d-c is applied.

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3.3.1.2 Solenoid F8208586.- Each solenoid shall meet the performance requirements of this specification when 208 volts, 60 cycles is applied.

3.3.2 High potential.- There shall be no puncture of insulation, no spark over, or decrease in dielectric strength of the insulation sufficient to cause failure of the solenoid, when subjected to the test specified in 4.2.1.

3.3.3 Insulation resistance. - The insulation resistance between the conductor and ground (case) shall be not less than 50 megohms, when subjected to the test specified in 4.2.2. The insulation resistance of solenoids subjected to the temperature rise test shall be measured immediately following that test and shall be not less than 5 megohms.

3.3.4 Temperature rise. - The temperature rise of the solenoid shall not exceed 75°C (135°F) after 2 minutes of continuous excitation with rated voltage applied. During tests the plunger of the a.c solenoid shall be fully extended (zero air gap), and the plunger of the d.c. solenoid shall be extended 7/8 inch (1/8 inch air gap).

3.4 Performance

3.4.1 Plunger force. - When the solenoid is energized (plunger extended) with rated voltage, it shall require a force of at least 100 pounds to push the plunger into the solenoid body. Test shall be conducted in accordance with 4.3.1.

3.4.2 Life cycle. - The solenoid shall not suffer mechanical or electrical damage after being subjected to 10,000 cycles of operation when tested in accordance with 4.3.2.

3.5 Environmental

3.5.1 Temperature. - The solenoid shall meet the requirements of 3.4.1 within an air temperature range extending from plus 51.1°C (+125°F) (minimum exposure of 4 hours with full impact of solar radiation 360 BTU/Ft. Sq/Hr.), to minus 31.2°C (-25°F) (minimum exposure of 3 days without benefit of solar radiation). The solenoid shall be capable of safe storage within the air temperature range extending from plus 71°C (+160°F) (for periods as long as 4 hours per day) to minus 62.2°C (-80°F) (for periods of at least 3 days duration). Test shall be conducted in accordance with 4.4.2.

3.5.2 Humidity. - The solenoid shall meet the requirements of 3.4.1 immediately after being subjected to the humidity test specified in 4.4.3.

3.5.3 Rain. - There shall be no damage to the solenoid when subjected to the rain test in accordance with 4.4.4.

3.5.4 Vibration. - The solenoid shall not suffer damage or fail to meet the requirements of 3.4.1 immediately after being subjected to the vibration test specified in 4.4.5.

3.5.5 Shock. - The solenoid shall not suffer damage or fail to meet the requirements of 3.4.1 immediately after being subjected to the shock test specified in 4.4.6.

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3.6 Workmanship. - Workmanship shall be in accordance with Specification 51-7.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General. - Unless otherwise specified, each solenoid shall be inspected for conformance with this specification and with Specification 51-7. Unless otherwise specified, standard ambient tests shall be performed at a temperature of 60°F to 90°F, at a barometric pressure of 28 to 32 inches of mercury, and a relative humidity not greater than 80 percent.

4.2 Electrical

4.2.1 High potential. - The solenoid shall be subjected to a potential of 900 volts rms (for solenoid assembly, D7665749) and 1000 volts rms plus twice the rated voltage (for solenoid, F8208586), with a frequency of 60 cycles per second for a period of one minute. Three of the first five and one out of every additional fifty or fraction thereof shall be subjected to the test voltage which shall be applied at a rate not exceeding 500 volts (rms) per second. For all other units the test voltage may be applied instantaneously for a period not less than 5 seconds or more than 30 seconds. Insulation resistance test shall be performed immediately after this test.

4.2.2 Insulation resistance. - The insulation resistance shall be measured by an approved method using a source of d-c potential of 500 volts.

4.2.3 Temperature rise. - Three of the first five solenoids and one out of every additional fifty or fraction thereof shall be tested. The solenoid shall be tested in still air and shall not be mounted in a manner that will cause heat dissipation through the mounting. Immediately after this test the high potential and the insulation resistance tests shall be repeated. The temperature rise shall be determined as follows:

(a) A stop watch shall be started at the instant the unit is stopped.

(b) Necessary bridge connections shall be made.

(c) A value of resistance which is slightly smaller than the resistance of the winding shall be preset into the bridge. Observe the null indicator of the bridge; and the instant the needle crosses the "zero" position, record the time and the value of the resistance set into the bridge. This procedure shall be followed until five values of resistance versus time after shutdown are recorded. These resistance values shall then be plotted against the corresponding time, and the curve extrapolated to zero time. The value of (R) at zero time shall be used in equation (2) in order to obtain "T". The value "T" shall then be used in equation (1) to obtain "T_r".

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(1) $T_r = T - T_a$

(2) $T = \frac{R}{r} (234.5 + t) - 234.5$

Where: T_r = Temperature rise in degrees centigrade.

T = Winding Temperature in degrees centigrade.

T_a = Ambient temperature in degrees centigrade at end of temperature rise test.

R = Hot resistance of winding in ohms.
r = Cold resistance of winding in ohms.
t = Temperature of winding in degrees centigrade when cold resistance measurement was made.

4.3 Performance

4.3.1 Plunger force. - The solenoid shall be mounted in a vertical position, plunger end up. The solenoid shall be energized and the amount of weight tending to force the plunger into the solenoid body shall be increased gradually until the plunger is forced in. The force required shall be in accordance with 3.4.1. "NOTE: Excessive currents may flow and cause damage to the coil winding when the air gap is increased, therefore, a circuit breaker or other protective device shall be inserted in series with coil winding."

4.3.2 Life cycle. - The quantity of solenoids to be tested shall be in accordance with instructions forming part of the contract or furnished by the contracting officer in connection with the contract. The unit shall be mounted in a vertical position (plunger end up) and in a manner to avoid heat dissipation through the mountings. The plunger shall have a 5 to 10 pound load restraining it when the Air gap is maximum; the load shall increase at a linear rate so that when the air gap is zero (plunger fully extended) the load shall be between 45 and 55 pounds. Each cycle shall be such that the time duration of the "off" period is three times the time duration of the "on" period. The time duration of one cycle shall be 4 seconds. At the completion of this test, the solenoid shall be tested for conformance with 3.4.1. Solenoids used for this test shall not be considered for Ordnance acceptance.

4.4 Environmental

4.4.1 General. - The environmental tests specified shall be limited to those contained in this paragraph. Three of the first five solenoids and one out of every additional hundred or fraction thereof shall be tested.

4.4.2 Temperature. - The solenoid shall be exposed to an air temperature of minus 62.2°C (- 80°F) for a period of 3 days. The air temperature shall then be raised to minus 31.2°C (-25°F) and after the internal temperature of the solenoid has reached equilibrium (approximately 4 hours), it shall be tested for conformance with 3.4.1. The air temperature shall be raised to plus 71°C (+ 160°F)

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and after the internal temperature of the solenoid has reached equilibrium at that temperature, it shall be allowed to remain at that temperature for a period of 4 hours. The air temperature shall be reduced to plus 51.5°C (≠ 125°F) and after the internal temperature of the solenoid has reached equilibrium at that temperature, the top surface shall be exposed to 360 BTU/Ft. Sq/Hr., for a period of 4 hours. The solenoid shall then be tested for conformance with 3.4.1.

4.4.3 Humidity. - A test chamber that is capable of being sealed shall be used; the source of heat shall be arranged in such a manner that the radiant heat will not fall upon the unit being tested. A solenoid that has been tested and found to be in conformance with the electrical and performance requirements of this specification shall be placed in the chamber, the temperature of which is adjusted to plus 46.1°C ≠ 1.1°C (≠ 115°F ≠ 2°F) and the relative humidity adjusted to 95% ≠ 5%. The solenoid shall remain under the above conditions for a period of 6 hours. The temperature of the chamber shall be readjusted to plus 26.7°C ≠ 1.1°C (≠ 80°F ≠ 2°F) and the unit shall be allowed to remain under this new condition for a period of 6 hours. The transitional periods shall be 2 hours. The rate of change during that period shall be uniform. The above cycle shall be repeated ten times. Immediately after this test the unit shall be tested for conformance with 3.4.1.

4.4.4 Rain. - The solenoid shall be mounted in the test chamber to simulate installation conditions. The rain test chamber temperature shall be maintained between 20°C and 30°C (68°F to 86°F) throughout the test period. A simulated rainfall of 4 ≠ 1 inches per hour shall be produced by means of a water spray nozzle of such design that the water is emitted in the form of small droplets rather than a fine mist. The temperature of the water shall be maintained between 11°C and 24°C (51.8°F to 75.2°F). The water spray shall be directed on the solenoid for a period of 2 hours in directions ranging from vertical to 45° from the vertical. The simulated rain shall be directed on all surfaces of the solenoid that are normally subjected to rain when the instrument is in its operating position. At the completion of this test the solenoid shall be tested for conformance with 3.4.1.

4.4.5 Vibration. - The solenoid shall be mounted rigidly using its own mounting provisions, to simulate field use. Each unit tested shall be vibrated successively along each of the three mutually perpendicular directions with accelerations and excursions as shown below. Applied frequency shall be changed at a uniform rate between the limits indicated; the rate of change shall not exceed one cycle per second.

Frequency limits	Excursion	Acceleration	Time (Each direction)
0 - 10 cps	0.060 inch	- - - -	1 hour
10 - 20 cps	0.030 inch	- - - -	2 hours
20 - 400 cps	- - -	4.5 g's	2 hours

4.4.6 Shock. - The solenoid shall be mounted rigidly, using its own mounting provisions, to simulate field use. A shock of 20 g's shall be imparted three times in each direction along each of the three mutually perpendicular directions, making a total of eighteen shocks. Each shock impulse shall have a time duration

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of 11 ± 1 milliseconds; the maximum g value shall occur at approximately 5.5 milliseconds and shall be within ± 10 percent. Filters used shall have cut-off frequencies selected to eliminate all harmonics above the fundamental, from the measured shock pulse.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing, and marking shall be in accordance with instructions forming a part of the contract or furnished by the contracting officer in connection with the contract.

6. NOTES

6.1 Ordering data. - Contracts and invitations for bids should specify the title, number, date of this specification, and type of packing.

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